



EXPLAINABLE · MULTI-ENGINE · DEV-FIRST

Secure smart contracts before they secure value.

SUBMITTED BY

Mohit Tiwari

Assistant Professor · Department of Computer Science & Engineering

Email: mohit.t.bvcoe@gmail.com

Bharati Vidyapeeth's College of Engineering · New Delhi, India

Team: CyberShield AI (Solo Innovation)

IDEATHON SUBMISSION INNOSTORM 2026

An explainable AI copilot that correlates analyzer findings with code context, supporting evidence, and remediation guidance.

CyberShield AI is an Explainable AI platform for secure Solidity smart contract development and blockchain security.

SCAN → INTERPRET → REVIEW

Human-in-the-loop

AI advises. Teams approve.

THE PROBLEM

Detection exists.
Interpretation is fragmented.

Fragmented
scanner outputs

Unprioritized
alerts across tools

Opaque
AI recommendations

Late-stage
security reviews

THE SOLUTION

One review layer for security evidence.

CyberShield AI links deterministic analyzer outputs with contextual AI reasoning, source references, and suggested remediation for developer review.

Why CyberShield AI?

Multi-tool security consensus

Records agreement across Slither, Mythril, and Solhint.

Explainable AI reasoning

Links risk interpretation to code and analyzer evidence.

Human-in-the-loop decision making

Developers review findings, suggestions, and risk disposition.

Developer-first workflow

Organizes analysis around review and readiness assessment.

FROM NOISE TO DECISION

A continuous security feedback loop

01

Raw findings

different formats · duplicate alerts

→

02

Correlated evidence

agreement · preconditions · context

→

03

Reviewed action

explain · suggest · review · assess

EXPECTED IMPACT

A more reviewable security workflow.

Evidence

findings linked to sources

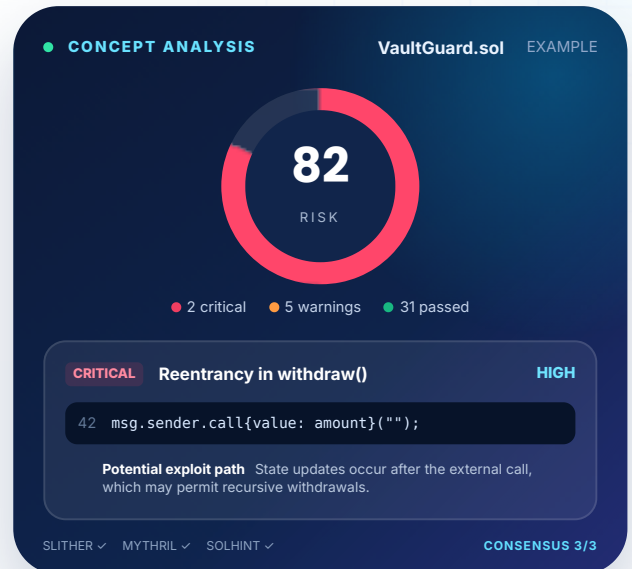
Priority

shared criteria for triage

Review

human-approved remediation

01 / 04



HOW IT WORKS

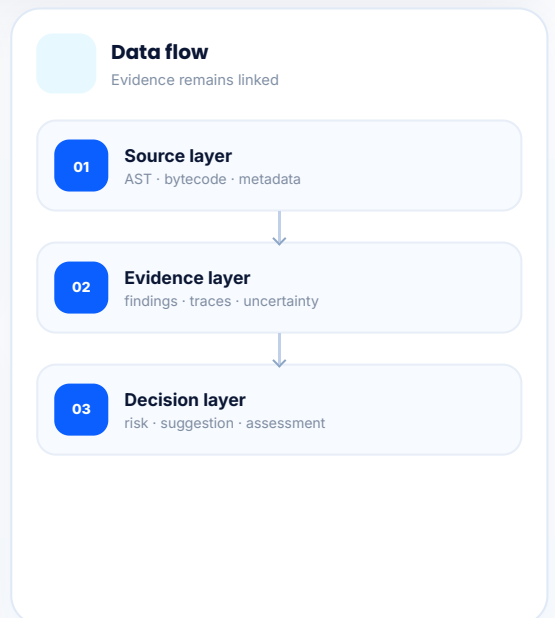
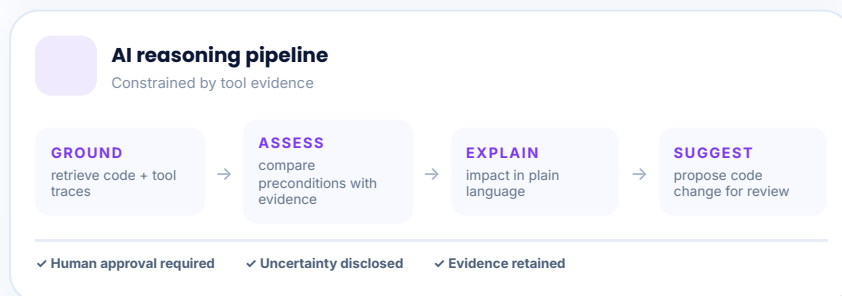
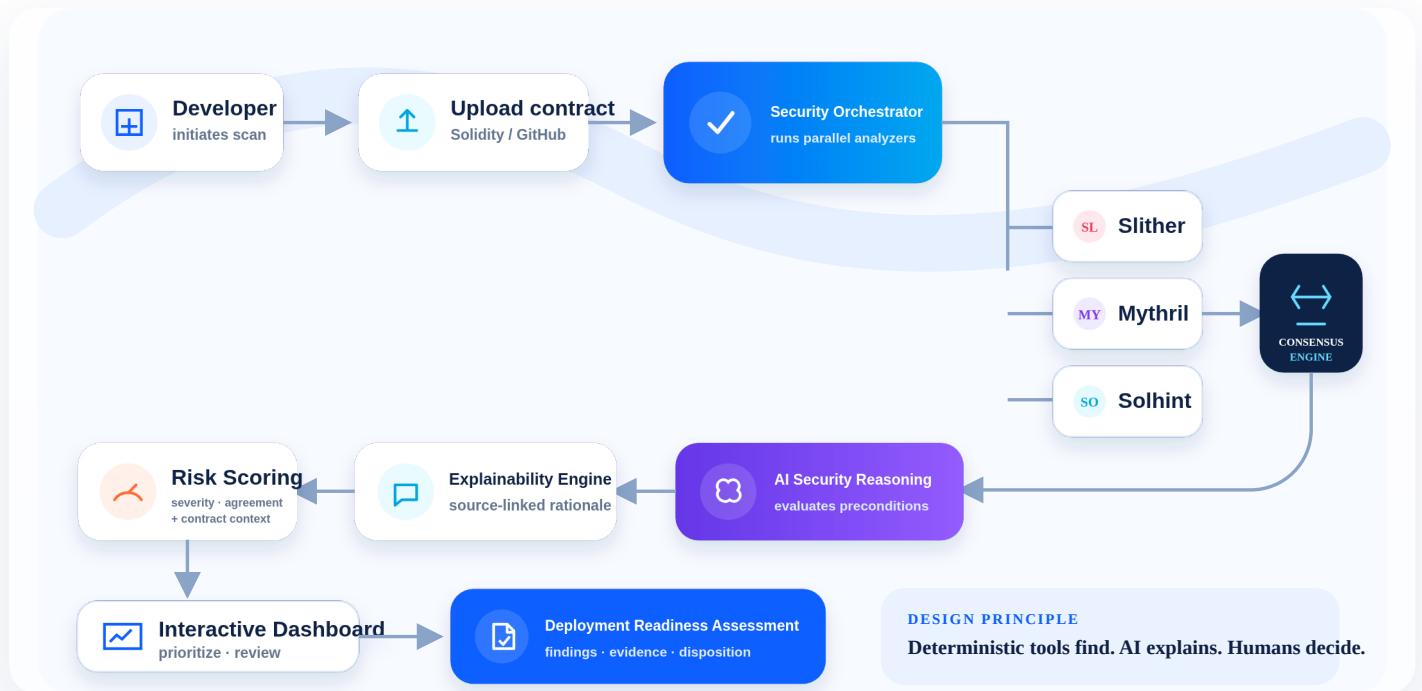
Deterministic detection.

Intelligent explanation.

The proposed system normalizes analyzer findings, records cross-tool agreement, and links each AI recommendation to source code and evidence.

REFERENCE ARCHITECTURE

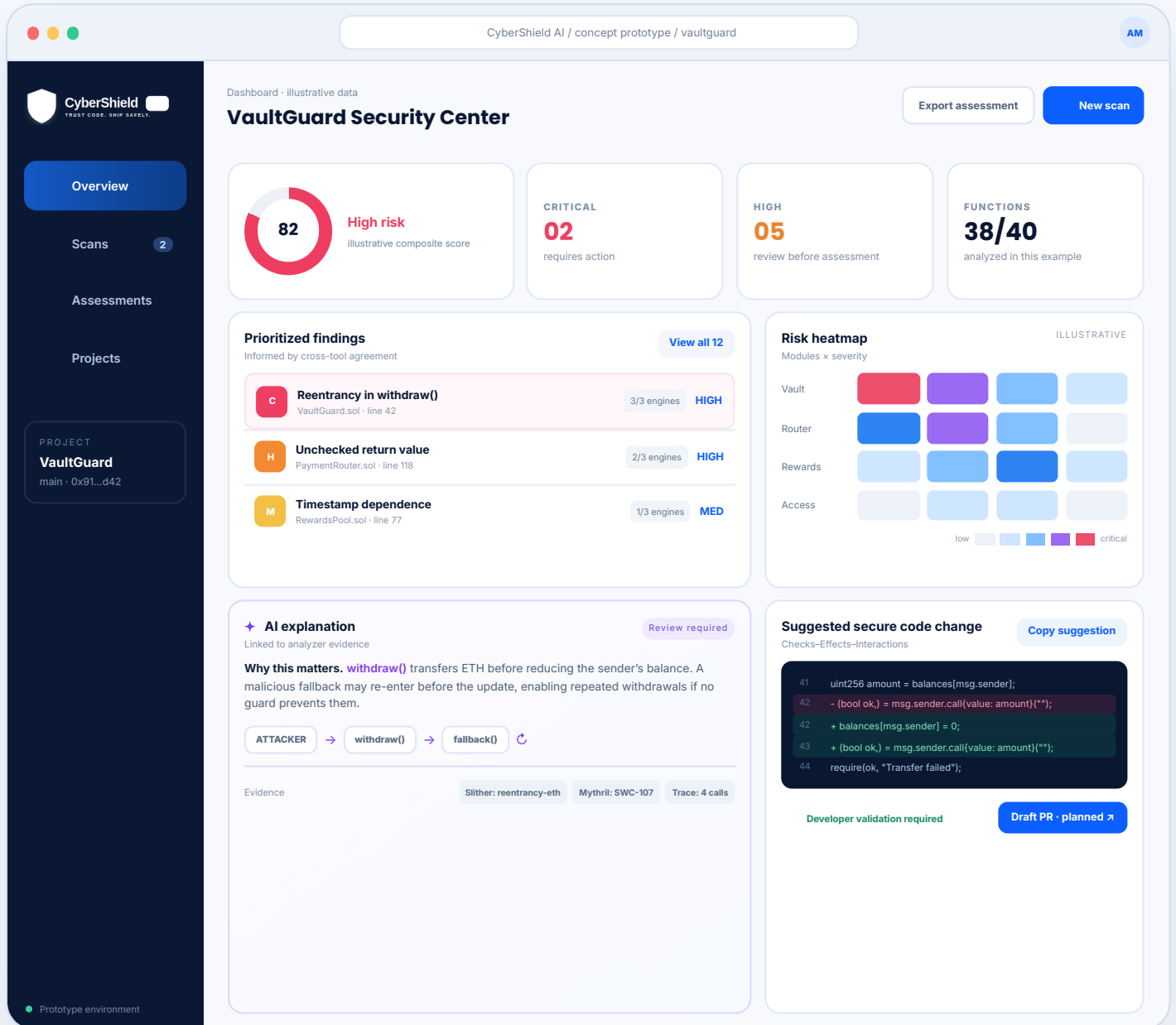
proposed · modular · evidence-linked



EXPERIENCE

Security evidence,
organized for review.

Concept interface showing how engineering, security, and release owners could inspect findings, evidence, and suggested remediation.



01Landing & onboarding

Project context and scan scope

02Upload contract

Solidity file · repository planned

03Readiness assessment

Evidence + ownership + disposition

04Explain & remediate

Risk context, reasoning, suggestion

EXECUTION PLAN

Implementation path.
Scope for expansion.

The proposal combines established open-source analyzers with a modular reasoning layer, explicit evidence storage, and staged developer validation.

TECHNOLOGY STACK

proposed implementation stack



React
Frontend



FastAPI
Backend



OpenAI or Gemini
LLM provider abstraction



Slither · Mythril · Solhint
Security engines



PostgreSQL
Evidence store



Docker
Containerized analyzer workers



GitHub
Version control + CI

INNOVATION HIGHLIGHTS

proposed differentiation

01

Consensus with provenance
Cross-tool agreement is recorded, not treated as proof.

02

Evidence-grounded AI
Reasoning begins with code and analyzer evidence.

03

Explanation as interface
Findings retain source references and rationale.

04

Readiness assessment
Outputs support an explicit release review.

COMPETITIVE ADVANTAGE

integration scope · not an audit replacement

Capability	Single analyzer	Point-in-time audit	CyberShield AI
Cross-tool correlation	Limited	Varies	Proposed
Traceable AI explanation	Limited	Analyst-led	Proposed
AI-assisted remediation	Varies	Analyst-led	Proposed
CI/CD readiness assessment	Varies	Point-in-time	Planned
Recorded human disposition	Varies	Common	Proposed

Proposed = architecture scope Planned = roadmap item Other tools vary by provider

COMMERCIALIZATION

applications + GTM hypotheses

BUSINESS MODEL
Freemium SaaS
Proposed tiers: Community → Team → Enterprise

→

GO-TO-MARKET
Pilot through developer tooling
GitHub app · IDE extension · audit partners (planned)

DeFi protocols

Token platforms

DAOs

Audit firms

Web3 enterprises

\$ / developer
Pro seats

\$ / scan volume
API usage

Annual contract
Enterprise

12-MONTH DELIVERY ROADMAP

discovery → evaluated product

01

PROPOSED · 0-8 WEEKS
Validate
Threat corpus · 3 analyzers · evidence schema

02

MONTH 3-4
Pilot
Reviewer agreement · explanation utility

03

MONTH 5-8
Productize
Teams · API · GitHub integration · billing

04

MONTH 9-12
Evaluate expansion
Enterprise controls · chain-specific benchmarks

ABOUT THE INNOVATOR

project lead

Mohit Tiwari is an Assistant Professor in the Department of Computer Science & Engineering at Bharati Vidyapeeth's College of Engineering, New Delhi. His work focuses on cybersecurity, blockchain security, explainable AI, secure software engineering, and AI-assisted developer tools. CyberShield AI extends his ongoing work toward improving secure smart contract development through explainable and evidence-driven security analysis.

FUTURE SCOPE

Multi-chain analysis · Continuous monitoring · Approval-gated remediation research · Private-model deployment

EXPECTED IMPACT

Broaden access to evidence-oriented smart-contract security workflows across teams with different levels of security expertise.